

SudokuSolver

User Manual — Manuel utilisateur

Version 1.1.1

A bilingual Sudoku solver, generator and tutor

Un solveur, générateur et tuteur de Sudoku bilingue français/anglais.
Solves any valid grid in milliseconds, generates puzzles by difficulty,
and explains each step the way a human would.

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1. Overview / Présentation

English. SudokuSolver is a desktop application that solves, generates and teaches Sudoku puzzles. It is written in Python with a PyQt6 graphical interface and runs on Windows, Linux and macOS. Everything is fully bilingual (French / English) and every widget has a tooltip in both languages.

Français. SudokuSolver est une application de bureau qui résout, génère et enseigne les grilles de Sudoku. Écrit en Python avec une interface graphique PyQt6, il fonctionne sous Windows, Linux et macOS. L'interface est entièrement bilingue, chaque contrôle dispose d'une infobulle français/anglais.

Key features / Fonctionnalités clés

- Fast solver with bitmask constraint propagation and MRV backtracking
- Solveur rapide par propagation de contraintes et backtracking MRV
- 8 human-style techniques with step-by-step bilingual explanations
- 8 techniques humaines avec explications pas-à-pas bilingues
- Puzzle generator with 4 calibrated difficulty levels
- Générateur 4 niveaux, chaque niveau calibré par nombre d'indices *et* technique maximale requise
- Interactive 9×9 grid, pencil marks, conflict highlighting
- Grille interactive, candidats, détection d'erreurs en temps réel
- Crash reports anonymised automatically (no usernames, no paths)
- Rapports de crash automatiquement anonymisés (pas de nom d'utilisateur ni de chemin)

2. Installation

2.1 Requirements / Prérequis

- Python 3.10 or newer / Python 3.10 ou plus récent
- `PyQt6` $\geq$ 6.5, `Pillow` $\geq$ 10.0
- ~80 MB free disk space for the virtual environment
- ~80 Mo d'espace disque pour l'environnement virtuel

2.2 Windows

The easiest way is to run `launch.bat` from the project folder. It will detect Python, create a local virtual environment in `%APPDATA%\SudokuSolver\venv`, install dependencies on first launch, then start the application silently (no console window).

```
launch.bat
```

If something goes wrong (corrupted venv, broken install, missing package), run the repair mode which wipes the venv and reinstalls everything cleanly:

```
launch.bat --repair
```

Tip / Astuce

Le launcher cherche Python dans plusieurs emplacements standard (%LOCALAPPDATA%\Programs\Python\..., C:\Python3xx, %ProgramFiles%), puis tente `py -3` (Windows Python Launcher), puis `python3` / `python` dans le PATH. Il évite l’alias du Windows Store qui déclenche une installation au lieu de lancer Python.

2.3 Linux / macOS

```
chmod +x launch.sh
./launch.sh
```

The virtual environment is created in `$XDG_DATA_HOME/SudokuSolver/venv` (or `~/.local/share/SudokuSolver/venv` as fallback).

2.4 Manual install / Installation manuelle

```
python -m venv venv
# Windows
venv\Scripts\activate
# Linux / macOS
source venv/bin/activate

pip install -r requirements.txt
python main.py
```

Warning / Attention

The project directory is portable across PCs (NAS-friendly): application data stays in the project folder, but the virtual environment is always local to each PC. Never copy the `venv` folder between machines.

Le dossier du projet est portable entre PC (compatible NAS) : les données applicatives restent dans le projet, mais le `venv` est toujours local à chaque machine. Ne jamais copier le dossier `venv` entre machines.

3. The interface / L’interface

3.1 Main window / Fenêtre principale

The window has four main areas:

1. **Top bar** — difficulty selector and “New puzzle” button.
Barre supérieure — sélecteur de difficulté et bouton “Nouvelle grille”.
2. **Grid** — the interactive 9×9 board.
Grille — le plateau 9×9 interactif.
3. **Action bar** — Hint, Next step, Solve, Check, Candidates, Reset.
Barre d’actions — Indice, Étape suivante, Résoudre, Vérifier, Candidats, Réinitialiser.
4. **Explanation label + status bar** below.
Zone d’explication et **barre d’état** en bas.

3.2 Keyboard / Clavier

Key / Touche	Action
Arrow keys / Flèches	Move selection / Déplacer la sélection
1–9	Enter digit / Saisir un chiffre
0, Delete, Backspace	Clear cell / Effacer la case
Click / Clic souris	Select a cell / Sélectionner une case

3.3 Cell states / États d’une case

- **Given** — original clue, cannot be edited, displayed in bold.
Donnée — indice original, non modifiable, en gras.
- **User entry** — digit you typed, displayed in a distinct colour.
Saisie utilisateur — chiffre que vous avez tapé, en couleur différente.
- **Conflict** — cell contents violate row/column/box rules, background in red.
Conflit — le contenu enfreint une règle ligne/colonne/bloc, fond rouge.
- **Candidates** — small digits showing the possible values, toggled by the “Candidates” button.
Candidats — petits chiffres montrant les valeurs possibles, activés par le bouton “Candidats”.

4. Features / Fonctionnalités

4.1 Generating a puzzle / Générer une grille

Pick a difficulty in the combo box and click **New puzzle**. The generator creates a grid with rotational symmetry, guaranteed unique solution, and calibrates the clue count and hardest technique required to match the target difficulty.

Level / Niveau	Clues / Indices	Hardest technique / Technique max
Easy / Facile	36–45	Hidden Single
Medium / Moyen	30–35	Hidden Pair
Hard / Difficile	26–32	Box / Line Reduction
Expert	22–28	X-Wing

Generation runs in a background thread so the interface never freezes, even if the generator needs to retry several times to hit the target difficulty.

4.2 Hint / Indice

Click **Hint** to see the next logical move. The engine picks the simplest applicable technique, highlights the pattern cells in green and the eliminated candidates in red, and explains *why* the move works in your language.

Tip / Astuce

If no technique applies, the engine falls back to a guess based on the backtracking solver. This is very rare for generated puzzles but can happen on imported puzzles that require advanced techniques not yet implemented.

4.3 Next step / Étape suivante

Same as Hint, but *applies* the step automatically. Chain several “Next step” clicks to watch the puzzle solve itself technique by technique.

4.4 Solve / Résoudre

Fills in the complete solution instantly. The fast solver runs in a background thread and is capped at 2 000 000 search nodes to guarantee no freeze even on pathologically hard input.

4.5 Check / Vérifier

Compares every user-entered digit against the stored solution. Wrong cells are highlighted in red, correct cells in green. Useful when you're not sure whether a deduction was valid.

4.6 Candidates / Candidats

Toggles the pencil-mark display. When on, every empty cell shows its current candidate set in small digits. When off, the cells stay clean — useful for a less cluttered view when you're sure of a region.

4.7 Reset / Réinitialiser

Clears all user entries and restores the puzzle to its generated state. The original clues are preserved.

4.8 File menu / Menu Fichier

- **New / Nouvelle** — generate a new puzzle at the current difficulty.
- **Open / Ouvrir** — load an 81-character puzzle string from a `.txt` / `.sdk` file. File size is capped at 10 KB.
- **Save / Enregistrer** — save the current grid as an 81-character string (dots for empty cells).
- **Quit / Quitter** — close the application.

4.9 Language menu / Menu Langue

Switch the entire UI between French and English at runtime. The default is auto-detected from your system locale.

4.10 Help menu / Menu Aide

- **How to play / Comment jouer** — a short in-app guide.
- **Solving techniques / Techniques de résolution** — a full bilingual catalogue of the 8 implemented techniques. See also the separate `guide_techniques.pdf` shipped with the project.
- **About / À propos** — version and credits.

5. Under the hood / Sous le capot ---

5.1 Solver architecture

The solver represents candidate sets as 9-bit masks ($1 \ll (d-1)$ for digit d). Every cell knows the bitwise intersection of the allowed digits for its row, column and box. Placing a digit updates the peers' candidate masks in constant time.

Search uses a Minimum Remaining Values (MRV) heuristic: pick the empty cell with the fewest candidates, try each candidate in turn, propagate, recurse. This is the same pattern used by Peter Norvig's well-known solver, extended with an explicit node cap to bound worst-case effort.

5.2 Human-style techniques

The human solver implements eight techniques, tried in order of increasing difficulty:

Technique	What it finds / Ce qu'elle détecte
Naked Single	Cell with only one candidate left
Hidden Single	Digit that fits in only one cell of a unit
Naked Pair	Two cells sharing exactly two candidates
Naked Triple	Three cells sharing three candidates
Hidden Pair	Two digits that fit in only the same two cells
Pointing Pair	Box candidates aligned on a single row/column
Box/Line Reduction	Line candidates confined to one box
X-Wing	2×2 rectangular elimination pattern

See `guide_techniques.pdf` for worked examples of every technique.

5.3 Generator

Generation starts from a random valid solution (shuffled backtracking), then removes clues symmetrically while verifying after each removal that the puzzle still has exactly one solution. The result is rated by counting clues and running the human solver to identify the hardest technique required. If the rating doesn't match the target difficulty, generation retries up to 40 times before falling back to the closest match.

5.4 Threading and anti-freeze

All long-running work (generation, solving) runs in dedicated `QThread` workers. The UI thread stays responsive and the window never freezes, even if the user clicks “New puzzle” several times in a row or imports a pathologically hard grid.

Worker lifecycle is handled defensively: when a thread finishes, its Python wrapper is cleared before Qt's native object is deleted, so subsequent operations never touch a dangling reference.

5.5 Crash reports / Rapports de crash

Uncaught exceptions are caught by a `sys.excepthook` handler that writes a JSON crash report into the project's `crash_reports/` folder. The report contains the traceback, Python version, OS and application version — *never* the username or home directory. Paths are rewritten to `~` and the username is replaced with `<user>`.

6. Troubleshooting / Dépannage

6.1 “Python not found” on Windows

The launcher couldn't find a working Python interpreter. Install Python 3.10 or newer from <https://www.python.org/downloads/> and **make sure to tick “Add Python to PATH”** during installation. Then re-run `launch.bat`.

6.2 “Installation failed” after Python upgrade

If you upgraded Python and the old venv is stale, run:

```
launch.bat --repair
```

This deletes the virtual environment and reinstalls everything cleanly.

6.3 Generator is slow on Expert

Expert puzzles that specifically require X-Wing are rare, so the generator may retry up to 40 times before falling back to a Hard-level puzzle with a slightly more constrained clue count. This is by design and the background thread keeps the UI responsive throughout.

6.4 I imported a puzzle and Solve says “no solution”

Either the imported grid has a typo (wrong digit in a given), or it already contains a conflict (check the red highlights), or it genuinely has no solution. The solver also bails out after 2 000 000 search nodes — this almost never triggers on valid 9×9 grids.

6.5 Where are my settings stored?

Everything stays inside the project folder (`crash_reports/`, `logs/`, puzzles history). The only thing outside is the virtual environment, which lives in `%APPDATA%\SudokuSolver\venv` on Windows or `$XDG_DATA_HOME/SudokuSolver/venv` on Linux/macOS for cross-PC portability.

6.6 The app crashed — what should I do?

Check the `crash_reports/` folder inside the project. The most recent JSON file contains an anonymised traceback you can attach to a GitHub issue. Usernames and home paths are replaced automatically, so it's safe to share.

7. FAQ

Can I play on a phone? / Puis-je jouer sur téléphone ?

No — this is a desktop application. For mobile, use any Sudoku app from the store.

Non — c'est une application de bureau. Pour le mobile, utilisez une application Sudoku dédiée.

Does it phone home? / Y a-t-il de la télémétrie ?

No. No network requests, no analytics, no telemetry. Crash reports are written locally and only shared if you choose to.

Aucune. Aucune requête réseau, aucune analytique, aucune télémétrie. Les rapports de crash restent locaux sauf si vous les partagez vous-même.

What Sudoku formats are supported?

81-character strings with digits 1-9 and any of `.`, `0`, `-`, space for empty cells. Line breaks are tolerated.

Why is the “Solving techniques” guide a separate PDF?

Because it's a reference you'll want to browse while solving. The main manual (this document) is the *how to use the app* guide; the techniques guide is the *how to solve Sudoku* reference.

8. Credits / Crédits

SudokuSolver is MIT-licensed. Thanks to Peter Norvig for the classic constraint-propagation solver design, and to the Sudoku solving community for decades of accumulated technique catalogues.

Issues and contributions welcome at:

<https://github.com/ARP273-ROSE/SudokuSolver>

